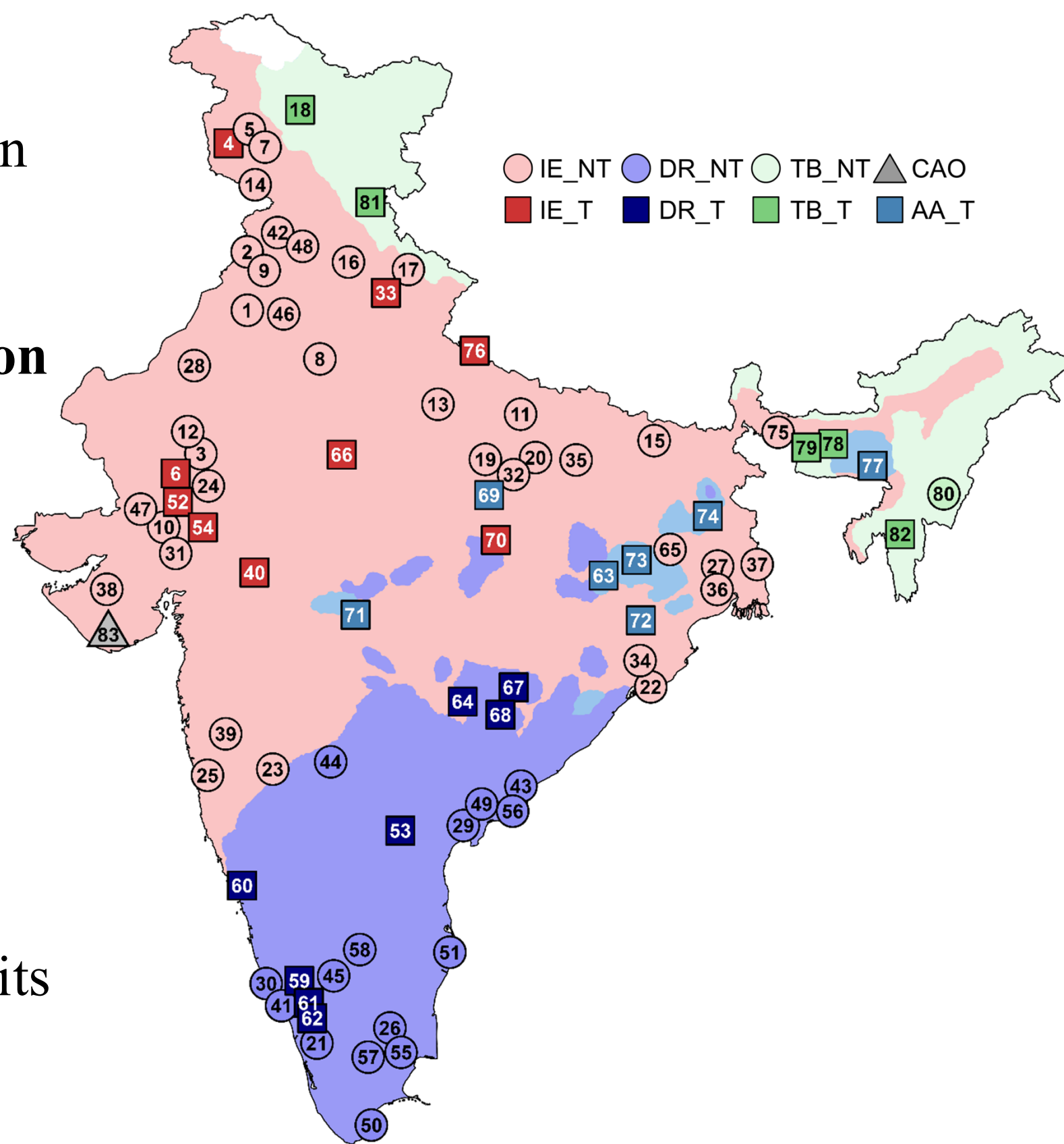


¹Ankit Mukherjee, ²Chandrika Bhattacharyya, ^{1,3}Mohammed Faruq

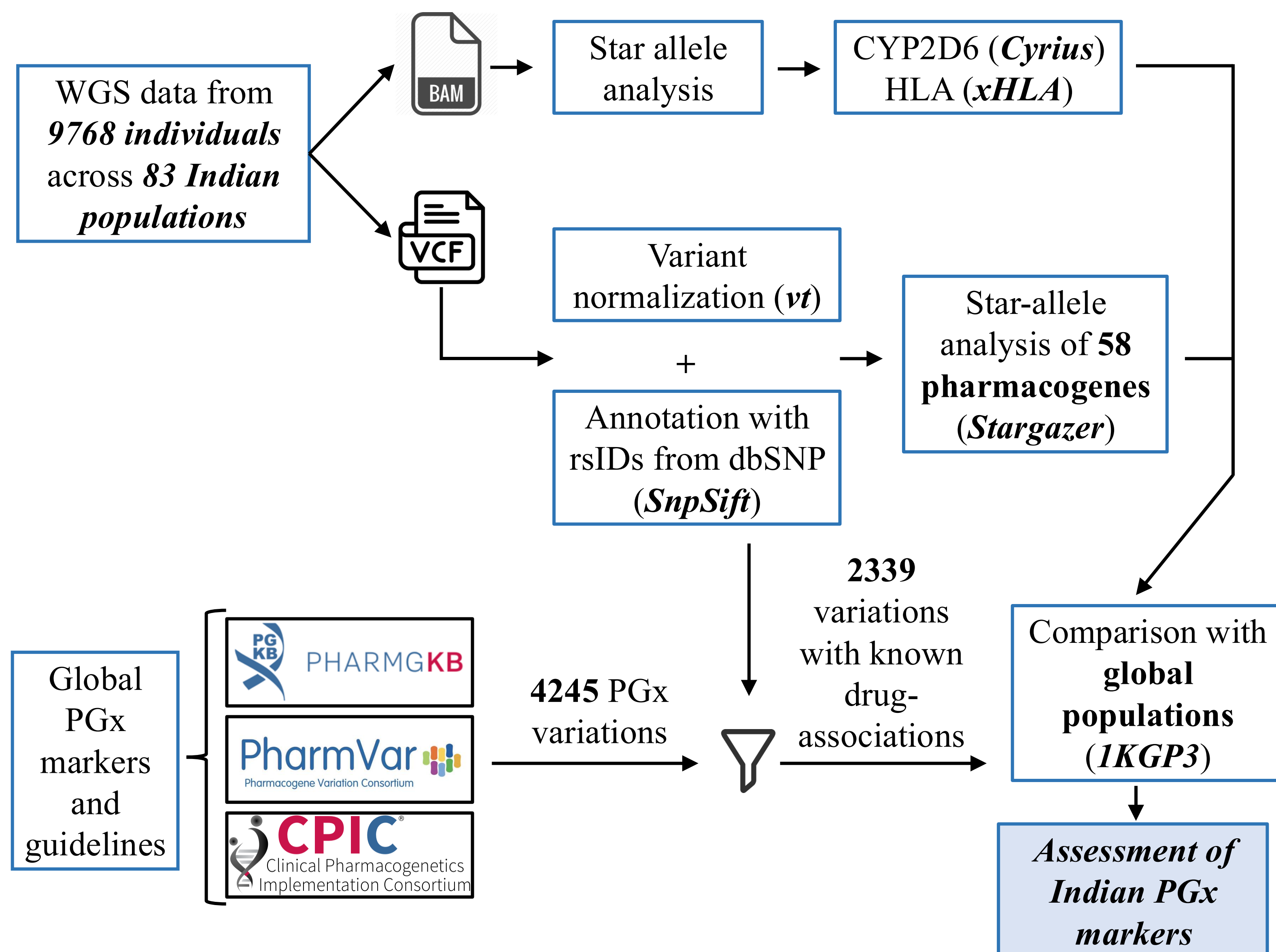
¹CSIR - Institute of Genomics & Integrative Biology (CSIR-IGIB), New Delhi, India, ²BRIC - National Institute of Biomedical Genomics (BRIC-NIBMG), Kalyani, India, ³Academy of Scientific and Innovative Research, Ghaziabad, India.

BACKGROUND

- Pharmacogenomic variation influences **drug efficacy, toxicity** and **dosage** requirements while populations can differ substantially in the frequency and distribution of clinically relevant alleles.
- India presents a unique landscape for pharmacogenomic implementation because of its extensive genetic and ethnolinguistic diversity.
- Previous genomic resources, including **IndiGen** and **GenomeAsia**, have substantially expanded the characterization of pharmacogenomic variation in South Asian populations. However, **most analyses have provided limited resolution of variations across India's diverse populations.**
- The **GenomeIndia** project, comprising **9768 individuals from 83 populations**, provides an unprecedented opportunity to characterize the population-level distribution of clinically actionable pharmacogenomic variation and evaluate its implications for precision medicine in India.

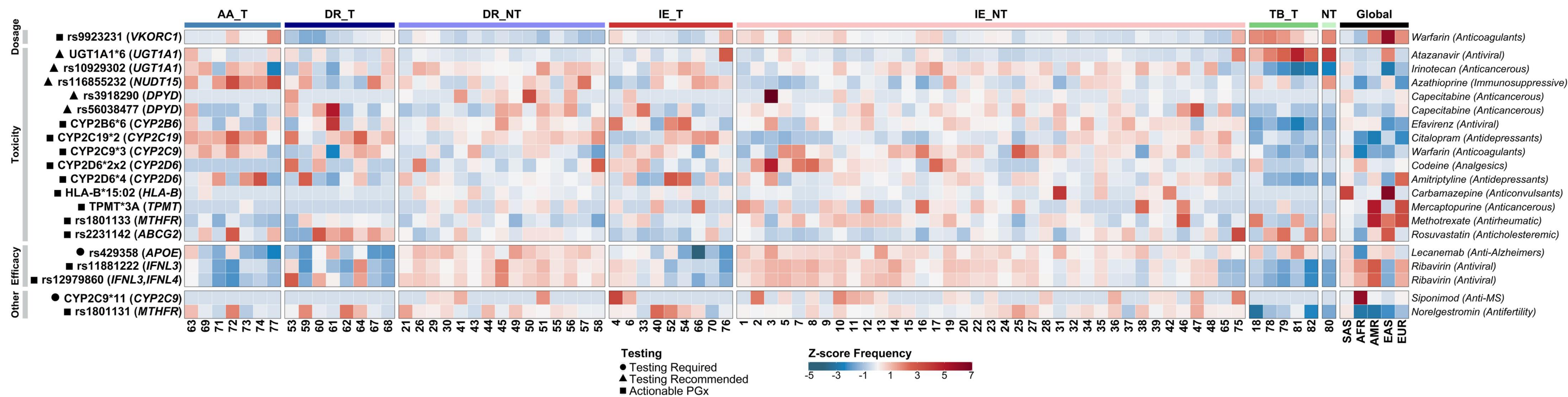


STUDY DESIGN

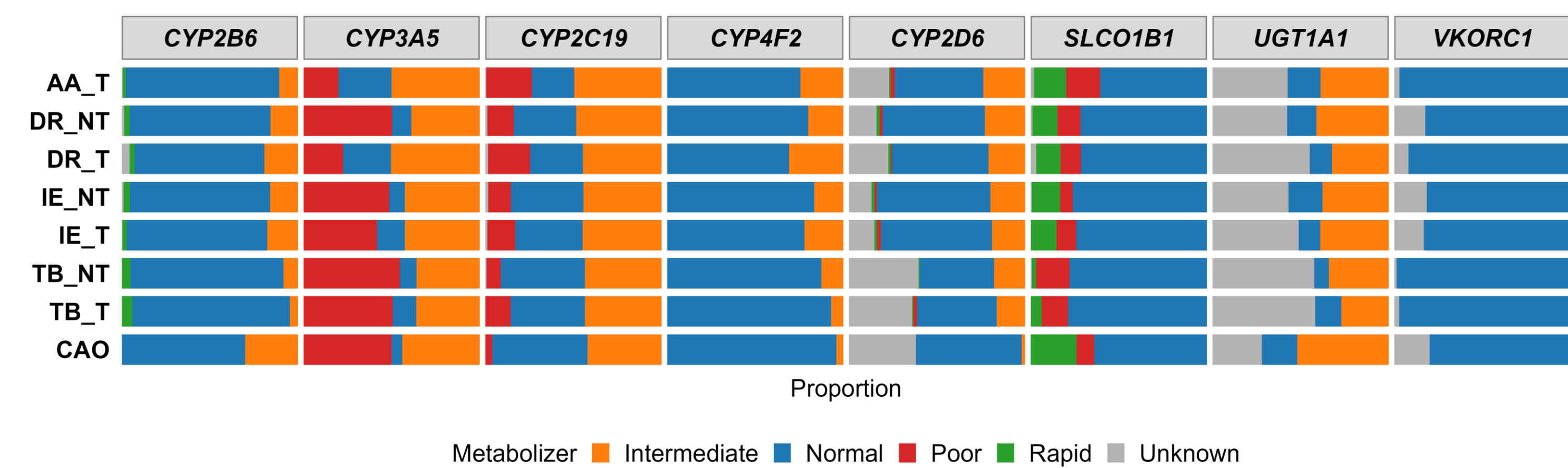


RESULTS

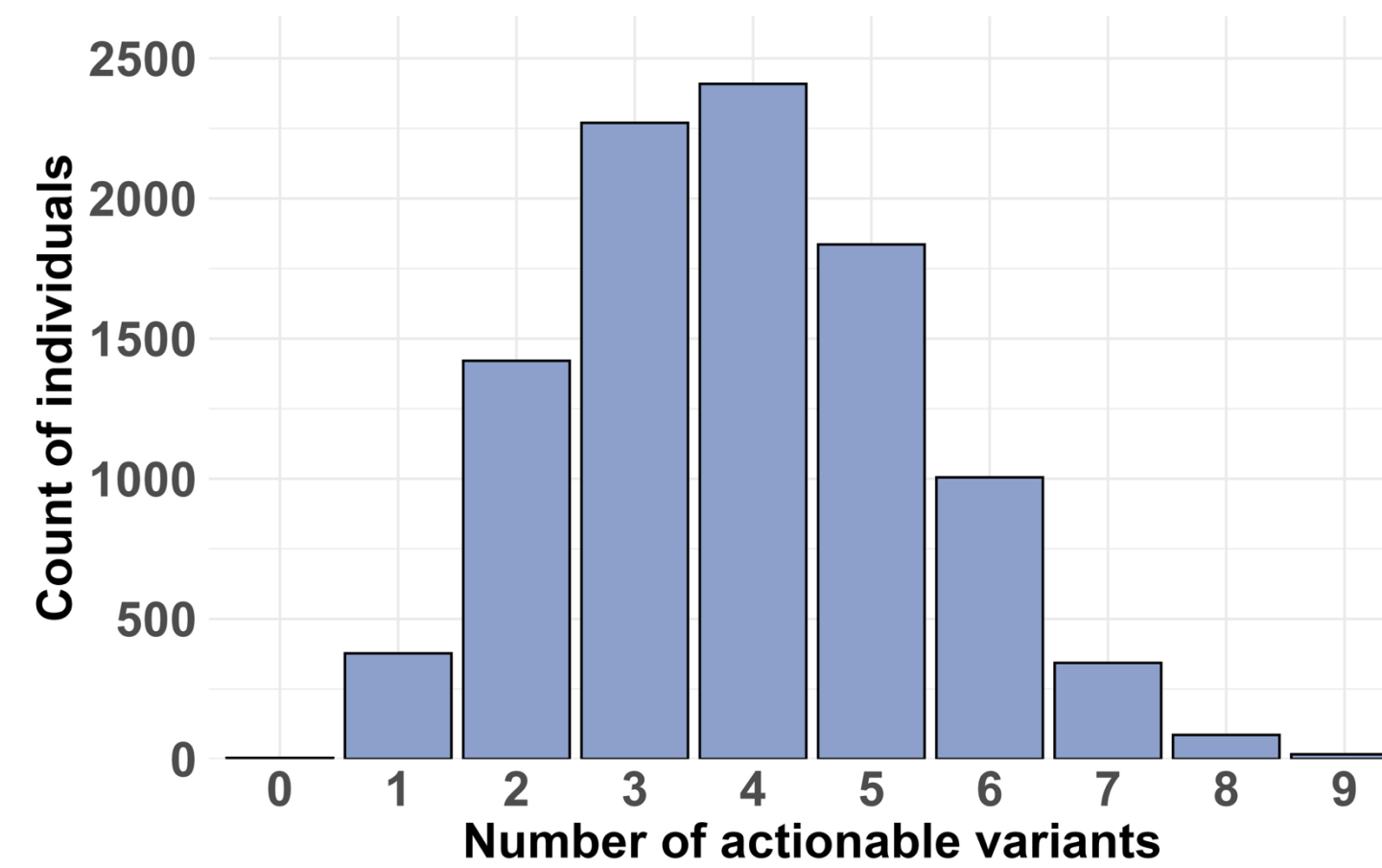
20 CLINICALLY ACTIONABLE PHARMACOGENOMIC VARIANTS



METABOLIZER PHENOTYPES



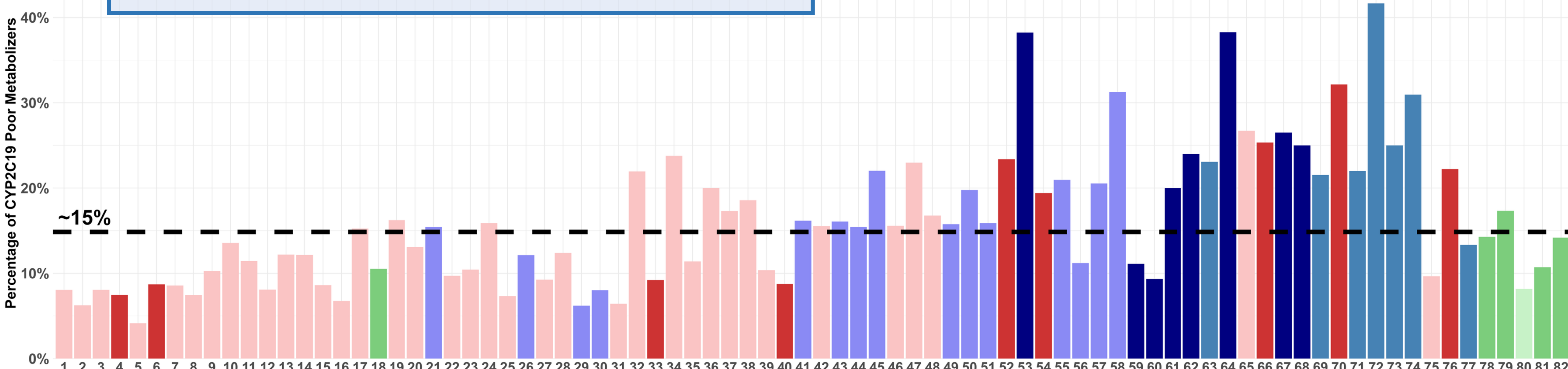
FOUR ACTIONABLE VARIANTS PER INDIVIDUAL



KEY THERAPEUTIC AREAS

Anticoagulants	Antineoplastics
<i>VKORC1</i> (Warfarin)	<i>DPYD</i> (Capecitabine)
Antivirals	Antidepressants
<i>UGT1A1</i> (Atazanavir)	<i>CYP2D6</i> (Amitriptyline)

CYP2C19 POOR METABOLIZERS



INCRETIN PATHWAY VARIANTS

Variant	Gene	Responder	% GI
rs10305420	<i>GLP1R</i>	Enhanced	31
rs6923761	<i>GLP1R</i>	Reduced	28
rs1800437	<i>GIPR</i>	Adverse	26

CYP2D6 STAR ALLELES IN GENOMEINDIA

Indian exclusive star alleles	Unknown Metabolizers %
*86, *111, *112, *113, *99	GenomeIndia = 18 Indigen = 9 SAS = 10

CONCLUSION

GenomeIndia reveals substantial population-level heterogeneity in clinically actionable pharmacogenomic variation that highlights the need for population-aware PGx strategies

Key pharmacogenes such as *CYP2C19*, *CYP2D6*, *VKORC1* and *DPYD* warrant deeper population-specific investigation and clinical validation in India

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